Training load and injury: part 1. The devil is in the detail – challenges to applying the current research in the training load and injury field

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Funding
No funding support has been obtained for the present manuscript.

Financial Disclosure and **Conflict of Interest.** The Authors affirm that they have no financial affiliation (including research funding) or involvement with any commercial organization that has a direct financial interest in any matter included in this manuscript

Authors' contribution
All the authors listed have contributed substantially according to ICMJE criteria: conception of the work in full or some sections, interpretation of published data, drafting the work or revising it critically for important intellectual content, approved the final version be submitted and published.

Data sharing
No original data were used for the commentary.

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Abstract

Background:
This article sets the scene for a critique of the research underpinning two common clinical assumptions: (1) training workload is a key factor influencing sports injury risk, and (2) it can be manipulated to reduce injury risk. In this summary we address why it is important for clinicians to critically evaluate the evidence behind research conclusions.
Clinical question:
Has research been designed and conducted well enough to help clinicians answer the questions '*what is the relationship between training workload and sports injury risk?' and 'can the metrics based on training workload be used to decrease the injury risk?'*
Key findings:
In the past decade, many sports injury researchers have developed new measures of exposure based on internal and external training workload to study the relationship between training load and injury. Some of these metrics may have been embraced by researchers and clinicians because (1) they are apparently supported by the scientific literature, (2) they are simple to calculate and use (averages and their ratio), and (3) there is an apparent reasonable rationale/narrative to support using workload metrics. However, (intentional or unintentional) questionable research practices and over-interpretation of research results undermines the trustworthiness of research in the training load-sports injury field.
Clinical application:
Clinicians should always aim to critically examine the credibility of the evidence behind a research conclusion before implementing research findings in practice. Something that initially looks promising and inviting might not be as revolutionary or useful as one first anticipated.

Key words: critical thinking, research quality, training load, injury

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Introduction

The relationship between training (work)load and sports injury is a prominent topic in sports science and medicine research. The hypothesis of a link between training load and sports injury is not new.[5] Manipulating training load using new measures of exposure (training load based metrics), assuming that changing load will cause the injury risk to reduce, has become common and recommended by international sports organizations.[8]

Clinical question

For many of us, the sole purpose of reading journal articles is to find information that will help us make better decisions in the clinic on Monday. If one reads something in *JOSPT*, or another scientific journal, supporting a relationship between training load and injury, should one accept the findings as true? In this article**,** we address why it is important for clinicians to critically evaluate the evidence behind conclusions in research (even when these recommendations are endorsed by international organizations), using the training load-injury field as an example.

Most popular training load metrics

The acute load, chronic load and their ratio (acute:chronic workload ratio, ACWR) are measures of exposure embraced by the scientific community and used as prognostic factors for injury risk. These measures of exposure are training load-based metrics that have been developed from a liberal interpretation of Eric Banister's model[6] proposed in the '70's to model training load (impulse) and physical performance. The time frames for acute and chronic workload, and the ACWR, were an approximation of the time decays derived from Banister's model.[6] The decays represented the time needed to dissipate the negative (fatigue) and positive (fitness) effects of training.

I**n** the current-day reinterpretation of Banister's model, acute load represented a surrogate measure of fatigue, and chronic load a surrogate measure of fitness. The two components of Banister's equation have been substituted by the ratio of two averages (rolling or exponentially-weighted moving averages): typically one week for the acute workload and four weeks for the chronic workload.[4] Acute workload, chronic workload and ACWR are the metrics one is most likely to see reported in research examining the association between training load and injury.

Training load metrics: make it as simple as possible but not simpler

It is not surprising new measures of exposure have gained popularity among clinicians and researchers. The metrics have been used in most studies, they are simple to calculate, easy to explain to athletes, and based on an apparently plausible rationale (do not train too much too soon) that fits commonly accepted training principles. However, these metrics are also an excessive simplification of complex injury etiology and of the mechanical load and damage training load generates. One need only take a short wade into the training load-injury field to finds copious and conflicting results regarding the relationship between training load and injury. It is difficult for clinicians to unravel the signal from the noise.[1]

Interpreting the results correctly and according to the nature of the study

Metrics including ACWR are based on Banister's model of performance, but have been shoehorned into injury research by attributing them a generic etiological role.[2] The associations between these measures of exposure and injuries are interpreted as supporting a causal link between load and injury (i.e. training load causes injury). Yet no studies have estimated any causal effects. Erroneous (causal) interpretation is not rare in sport medicine.[9]

Clinicians should always consider whether research results provide evidence of causal effects before deciding to change a prognostic factor (e.g. training load) in an attempt to alter the likelihood of an (adverse) event. The interpretation should be coherent with the nature of the study (descriptive, predictive or causal) and its limitations.

Questionable research practices: key barriers to trustworthy research results

Questionable research or reporting practices (QRPs) are problems with research design, analysis and reporting that impinge on the trustworthiness of the results.[3] Some QRPs can have very little effect on research results. Other QRPs can be fatal – obliterating the reader's trust in the results (Table 1).[7] Hypothesis with unsupported causality, causation claimed/assumed without proper designs or without discussing bias, and ignoring non-significant results and contradictory evidence, are examples of QRPs that blight training load-injury studies.

*** Table 1 ***

Research involves specific steps (Figure 1). At each step, there are threats to reproducible science (i.e. trustworthy research) such as p-hacking and HARKing (Hypothesizing After the Results are Known). It is important to consider how far QRPs might have infiltrated the training load-injury research field, and the implications for clinicians aiming to use research in their practice.

*** Figure 1 ***

Summary

In this article, we outlined why it is important to carefully assess the trustworthiness of research one might intend to use in practice. In part 2, we highlight various issues related to concepts and methods in the training load-injury research field. We focus on decisions researchers can make about hypotheses, study designs, and conducting, analyzing and reporting research,[10] how those decisions can impact on the trustworthiness of research results.

We aim to raise potential challenges to applying the current research in the training load and injury field, and identify some biases, to improve future research. We hope our commentary encourages clinicians to be careful when applying research in practice.

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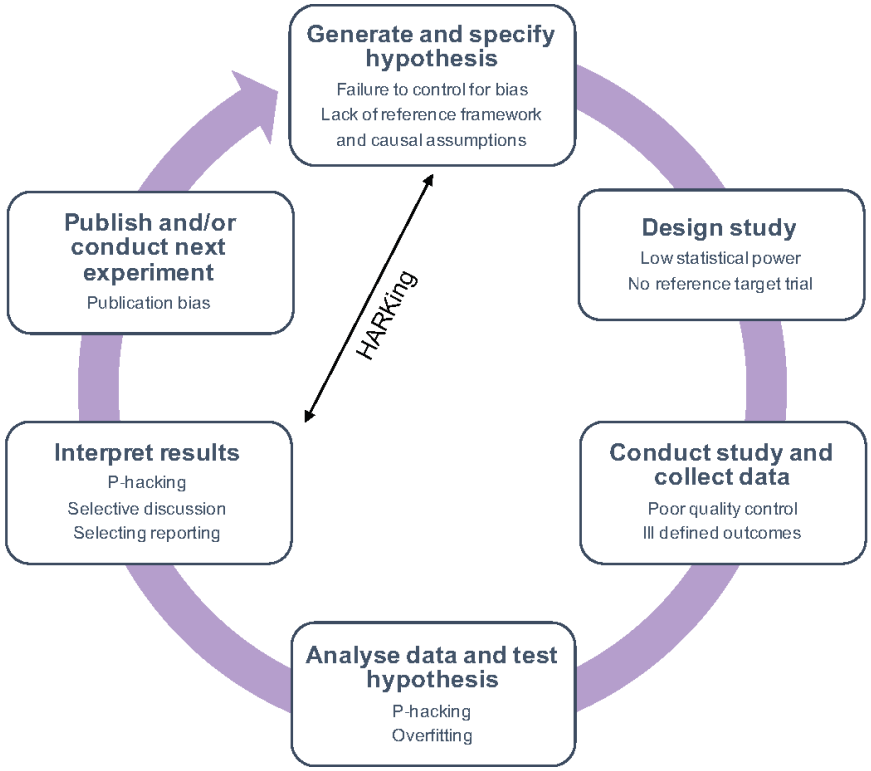

Figure 1. Potential threats to the scientific process. Abbreviation: HARKing, hypothesizing after the results are known. Adapted with permission from Munafò et al.[7] under a Creative Commons Attribution 4.0 International License (http://creativecommons.org/licenses/by/4.0/).

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Table 1. List of common questionable research practices (modified from Gerrits et al.)

Questionable research practices
1. Non significance of results is not addressed
2. Causation claimed without appropriate design or analysis
3. Hypothesis contains unsupported causality
4. Potential causal relationship is not justified
5. Causation is claimed without discussing bias
6. Inappropriate use of evidence
7. Generalisation to different population, setting or location
8. Secondary outcomes are overstated
9. Clinical relevance of non-significant results
10. Small effect size is overstated
11. Precision of estimated not discussed or considered
12. Outcome measure does not reflect objectives
13. Supporting literature based on same underlying data
14. Abstract does not reflect the main findings
15. Title does not reflect the main findings
16. Order of discussion differs from aim
17. Objectives are phrased differently in discussion
18. Generalisation is not supported by sample
19. Results section contains interpretation
20. Conclusion does not reflect outcome measure
21. Objectives are not reflected in the discussion
22. Conclusions in abstract do not reflect main text
23. Limitations are poorly discussed
24. Supporting evidence is poorly documented
25. Objective is not reflected by the conclusions
26. Conclusions do not reflect findings in context
27. Impact of limitations on results not discussed
28. Conclusions do not reflect findings
29. Contradictory evidence is not mentioned
30. Recommendations for practice are lacking or not supported by the findings
31. Implications for practice are lacking or not supported by the findings

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